Insectivorous Mammal Molar from the Miocene Bukpyeong Formation, Donghae City, South Korea

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In 2004, several microvertebrate teeth were collected from the Miocene Bukpyeong Formation by the wet screening of fossiliferous sediments at an outcrop exposed in Jiga-dong, Donghae-si, South Korea. Here, we describe an unidentified mammal tooth, specifically a right lower molar from an insectivorous mammal. The crown is well preserved with a maximum length of 2.2 mm and a width of 1.7 mm. It displays typical tribosphenic molar characteristics with the protoconid and hypoconid being the tallest cusps in the trigonid and talonid, respectively. In the talonid, the entoconid and hypoconid are connected by a postcristid, which separates the hypoconulid laterally. The overall morphology closely resembles the myotodonty structure of a bat molar, suggesting a potential affinity with the family Vespertilionidae. However, the absence of a distinct cingulum, a feature present in most chiropteran molars, leaves open the possibility that it may belong to another insectivorous mammal group, such as the Eulipotyphla. This discovery is significant as it provides evidence of mammalian faunal diversity in the Miocene of South Korea and represents the first record of an insectivorous mammal from this period in the region.